

LAB 2

Exercise 2: Internet Tools, Email Services and Cyber Hygiene (CO1)

Students should use internet services effectively for communication and information access, configure email services, and follow safe cyber practice

PART A — INTERNET BROWSING

Task 1: Introduction to Internet Browsing

Objective: To understand web browsers and perform basic internet browsing using Google Chrome or Mozilla Firefox.

Tools Required

- Computer/Laptop
- Internet connection
- Google Chrome / Mozilla Firefox

Theory

A **web browser** is a software application used to access, retrieve, and view information on the World Wide Web. When you type a web address (URL), the browser contacts a web server, downloads the site's raw files (like HTML and CSS), and translates them into the readable pages, text, and images you see on your screen.

Examples:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari
- Opera

A browser uses a **URL (Uniform Resource Locator)** to locate a website.

Example:

`https://www.google.com`

Important Browser Components

Component	Function
Address Bar	Enter website address or search query
Back	Return to previous page
Forward	Move to next page
Refresh	Reload current webpage
Home	Open home page

New Tab	Open another webpage
Bookmark	Save a webpage
History	View previously visited websites
Downloads	View downloaded files

Procedure

1. Open Google Chrome.
2. Locate the address bar.
3. Type `https://www.google.com`.
4. Press **Enter**.
5. Open a new tab using `Ctrl + T`.
6. Visit another website.
7. Use the **Back** and **Forward** buttons.
8. Refresh the webpage using `Ctrl + R`.
9. Bookmark an important webpage using `Ctrl + D`.
10. Open the browser History using `Ctrl + H`.
11. Open the Downloads section using `Ctrl + J`.



Practice Example

Task: Find the official website of the Government of India.

Student have to write the result?

PART B — INTERNET SEARCH TECHNIQUES

Task 2: Effective Internet Searching

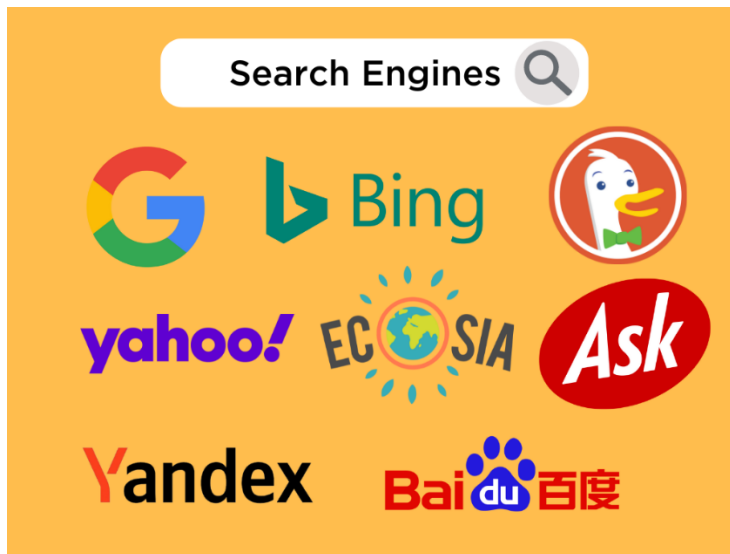
Objective: To learn different techniques for obtaining accurate information from search engines.

Theory

A **search engine** is a software program or information retrieval system designed to help users locate information stored on the World Wide Web or within a specific database.

Examples:

- Google
- Bing
- DuckDuckGo
- Yahoo



Simply entering a long sentence into a search engine may produce unnecessary results. Search operators can make searches more precise.

Common Search Techniques

1. Exact Phrase Search

Use quotation marks.

Example:

"digital literacy in education"

This searches for the exact phrase.

2. Excluding a Word

Use the minus – operator.

Example:

`apple benefits -iphone`

This searches for information about apples while excluding results related to iPhones.

3. Site-Specific Search

Use:

`site:`

Example:

`site:gov.in digital education`

This searches mainly within government websites.

4. File Type Search

Use:

`filetype:`

Example:

`cyber security awareness filetype:pdf`

5. Searching a Particular Website

Example:

`site:ugcnet.nta.ac.in UGC NET syllabus`

Practice Exercise

Search for:

1. `"cyber hygiene"`
2. `site:gov.in cyber security`
3. `cyber security filetype:pdf`
4. `digital literacy -social media`
5. `"Google Drive" tutorial`

Practice Example

Question: Find a PDF about cyber security published on a government website.

Student should write result of their search.

PART C — EMAIL SERVICES

Task 3: Creating and Configuring a Gmail Account

Objective: To create and configure an email account and understand the basic components of email.

Tool: Gmail

Theory

An **email** (electronic mail) is a digital messaging system used to transmit text, files, images, and videos over a computer network. It requires an **email account** to act as a digital mailbox, allowing you to send messages instantly to anyone across the world using their unique email address.

A typical email address has the structure:

`username@domain.com`

Example:

student123@gmail.com

Important Email Fields

Field	Meaning
To	Main recipient
Cc	Carbon Copy
Bcc	Blind Carbon Copy
Subject	Short description of email
Attachment	File sent with email
Body	Main message

Procedure

1. Open a web browser.
2. Open Gmail.
3. Select **Create account** if an account is required.
4. Enter the required information.
5. Choose a unique email address.
6. Create a strong password.
7. Complete account verification.
8. Sign in to Gmail.
9. Explore Inbox, Sent, Drafts, Spam and Trash.

Practice Example

Create an email account for academic use.

Example username:

`student.name2026`

Avoid using easily guessable information such as:

`student123456`

or

`password123`

Important Rule

Never share your Gmail password, OTP or recovery codes with another person.

PART D — EMAIL COMPOSITION AND MANAGEMENT

Task 4: Sending an Email

Objective: To compose and send an email with an attachment.

Procedure

1. Log in to Gmail.
2. Click **Compose**.
3. Enter the recipient's email address.
4. Enter an appropriate subject.
5. Type the message.
6. Attach a document using the attachment icon.
7. Verify the recipient and attachment.

8. Click **Send**.

Practice Example

Situation: You have completed a practical assignment and need to submit it to your teacher.

Example subject:

Computer Lab Assignment – Internet Tools

Attach a PDF or Word document.

Student Activity

Send an email to the instructor with:

- Appropriate subject
 - Short professional message
 - Assignment attachment
-

Task 5: Cc, Bcc, Reply and Forward

Objective: To understand different email communication options.

Cc

Cc means **Carbon Copy**.

Recipients in Cc can see the other recipients.

Bcc

Bcc means **Blind Carbon Copy**.

Recipients in Bcc generally cannot see the other Bcc recipients.

Reply

Used to respond to the sender.

Reply All

Used when the response needs to reach all relevant recipients.

Forward

Used to send an existing email to another person.

Practice Example

Question: Student should send mail to their friends using above features.

Task 6: Email Organization and Management

Objective: To organize emails efficiently.

Gmail Features to Practice

- Labels
- Archive
- Star
- Mark as unread

- Search
- Delete
- Spam reporting
- Filters

Procedure

1. Open Inbox.
2. Select an academic email.
3. Apply a label such as **College**.
4. Star an important email.
5. Archive an old email.
6. Search for an email using keywords.
7. Move an unnecessary email to Trash.
8. Check the Spam folder.

Practice Search

Search Gmail using:

`assignment`

Then try:

`from:teacher@example.com`

and:

`has:attachment`

Students should observe how search filters help locate messages.

PART E — CLOUD STORAGE

Task 7: Google Drive

Objective: To learn how to store and manage files using cloud storage.

Theory

Cloud storage is a service model where data is transmitted, managed, and backed up remotely over the internet. Instead of saving files exclusively on a local hard drive, information is stored on virtual servers maintained by a third-party cloud service provider.

Examples include:

- Google Drive
- OneDrive
- Dropbox
- iCloud

Google Drive Operations

Students should practice:

1. Creating a folder.
2. Uploading a file.
3. Creating a Google Doc.
4. Creating a Google Sheet.
5. Renaming a file.
6. Moving a file.
7. Downloading a file.
8. Sharing a file.
9. Managing permissions.

Procedure

1. Open Google Drive.
2. Click **New**.
3. Create a folder named:

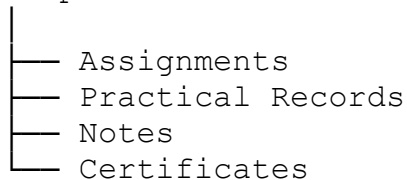
Computer Lab 2026

4. Open the folder.
5. Upload a PDF.
6. Rename the file.
7. Create a Google Doc.
8. Enter five lines of text.
9. Rename the document.
10. Download a copy.

Practice Example

Create the following folder structure:

Computer Lab 2026



Upload at least one file into each appropriate folder.

Expected Result

Students should be able to organize academic files using cloud storage.

Task 8: File Sharing and Permissions

Objective: To understand secure file sharing through Google Drive.

Common Permissions

- Viewer
- Commenter
- Editor

Procedure

1. Select a file in Google Drive.
2. Click **Share**.
3. Add another user's email address.
4. Select the required permission.
5. Send the sharing invitation.
6. Change the permission.
7. Remove access.

Practice Example

Create a document titled:

Cyber Hygiene Practical

Share it with a classmate as **Viewer**.

Then change the permission to **Editor**.

Finally, remove the classmate's access.

Important Cyber Hygiene Rule

Do not select **Anyone with the link** for confidential documents unless there is a genuine need.

PART F — ONLINE COMMUNICATION TOOLS

Task 9: Online Communication and Collaboration

Objective: To understand responsible use of online communication tools.

Examples

- Google Meet
- Microsoft Teams
- Zoom
- Google Chat
- WhatsApp Web
- Other approved communication platforms

Activities

Students should practice:

1. Joining an online meeting.
2. Muting/unmuting microphone.
3. Turning camera on/off.
4. Using chat.
5. Sharing screen where permitted.
6. Leaving a meeting.
7. Understanding meeting invitations.

Practice Example

Conduct a mock online class.

One student acts as the teacher and others as students.

Practice:

- Joining the meeting.
- Muting microphone.
- Sending a chat message.
- Sharing screen.
- Leaving the meeting.

Cyber Safety

Never share an online meeting link publicly if it is intended for a private class or organization.

PART G — CYBER HYGIENE

Experiment 10: Understanding Cyber Hygiene

Objective: To identify common cyber threats and practice safe internet behavior.

What is Cyber Hygiene?

Cyber hygiene refers to regular practices that protect computers, accounts, personal information and digital identity from cyber threats.

Antivirus:

Antivirus software is a program that either comes installed on your computer or that you purchase and install yourself. It protects your computer against most viruses, worms, Trojan horses and other unwanted invaders that can make your computer sick.

Firewall:

A firewall is a special software or hardware designed to protect a private computer network from unauthorized access. A firewall is a set of related programs located at a network gateway server which protects the resources of the private network from users from other networks.

PROCEDURE:

- Installing Symantec antivirus for Windows:
- Insert Symantec antivirus CD into your CD drive
- Double click on the Symantec-setup.exe
- The installer will open
- Click next to proceed
- License agreement will open. Click I accept the terms of the
- License agreement and then click next.
- Follow the instruction on the screen to complete the installation.
- Get Computer Updates:
- Click start> settings>control panel
- Click Automatic Updates icon to open Automatic Updates dialog box
- Check the box Keep my computer up to date
- Choose a setting
- Click OK

Block Pop ups:

- In the IE open tools>pop-up blocker
- Go to Start>control panel>Network and Internet Connections >windows firewall
- In the general tab check the On(recommended) box
- If you don't want any exceptions check on Don't allow exceptions box

Major Cyber Threats

1. Phishing

Fraudulent messages designed to trick users into revealing:

- Passwords
- OTPs
- Bank details
- Personal information

2. Malware

Malicious software that can damage or compromise a device.

Examples:

- Virus
- Trojan
- Worm
- Ransomware
- Spyware

3. Social Engineering

Manipulating people into revealing information or performing unsafe actions.

4. Fake Websites

Websites designed to look like legitimate websites but created to steal information.

5. Identity Theft

Unauthorized use of another person's personal information.

Core Checklist for Personal Cyber Hygiene:

To protect your personal data and digital life, integrate these essential habits into your daily or weekly routines:

- **Password Security:** Use unique, random passwords of at least 14 to 16 characters for every single account. Never reuse credentials across platforms, and deploy a trusted **password manager** to store them safely.
- **Multi-Factor Authentication (MFA):** Turn on MFA or Two-Factor Authentication (2FA) for all critical accounts. This provides a vital layer of identity verification beyond a standard password.
- **Automated Updates:** Enable automatic updates on your operating systems, mobile applications, and internet browsers. Regular patching fixes dangerous security vulnerabilities before hackers can exploit them.
- **Data Backups:** Store copies of important documents and photos on an encrypted cloud service or an offline external hard drive. Routinely check that your recovery process works smoothly.
- **Skepticism Online:** Inspect links and sender addresses before clicking or downloading attachments. Avoid public, unsecured Wi-Fi networks for financial transactions unless using a trusted VPN

Task 11: Identifying a Phishing Email

Objective: To identify suspicious emails and messages.

Example

Suppose a student receives:

Subject: Your Account Will Be Closed Today!

Dear Customer,

Your account will be suspended immediately. Click here to verify your password and OTP.

Identify Warning Signs

1. Urgent or threatening language.
2. Request for password.
3. Request for OTP.
4. Suspicious link.
5. Unknown sender.
6. Spelling/grammar mistakes.
7. Unusual attachment.

Student Activity

For each suspicious message, answer:

Question	Answer
Is the sender known?	Yes/No
Is there an urgent demand?	Yes/No
Is a password requested?	Yes/No
Is OTP requested?	Yes/No
Is there a suspicious link?	Yes/No
Is an attachment unexpected?	Yes/No
Should you click?	Yes/No

Correct Action

Do not click suspicious links or attachments.

Verify the information through the organization's official website or another trusted channel.

PART H — PASSWORD MANAGEMENT

Task 12: Creating Strong Passwords

Objective: To understand secure password practices.

Characteristics of a Strong Password

A strong password should ideally:

- Be long.
- Be unique.
- Not contain easily guessed personal information.
- Not be reused across important accounts.
- Use a combination of different character types where appropriate.

Weak Password Examples

Avoid passwords such as:

123456
password
qwerty
ravi123
india123

Better Approach: Use a long, unique passphrase that is difficult for others to guess.

Password Manager: A password manager can securely store unique passwords so that users do not need to memorize every password.

Student Activity

Create sample passwords **for practice only** and classify them as:

- Weak
- Moderate
- Strong

Experiment 13: Multi-Factor Authentication

Objective: To understand the importance of additional account security.

Theory

Multi-Factor Authentication (MFA) requires more than one form of verification.

Examples:

1. Password
2. OTP
3. Authenticator application
4. Security key
5. Biometric verification

Practice

Using a training/demo account where appropriate:

1. Open account security settings.
2. Locate two-step verification/MFA.
3. Understand the available verification methods.
4. Enable MFA if permitted.
5. Review recovery options.

Important

Never share an OTP with another person, even if the person claims to be from a bank, company, university or government organization.

PART I — SAFE INTERNET USAGE

Task 14: Safe Internet Practices

Objective: To develop safe browsing and online behavior.

Safety Practices

Before Clicking a Link

Check:

- Who sent it?
- Is the message expected?
- What website does the link lead to?
- Is the domain correct?
- Is the request reasonable?

While Browsing

- Keep browser updated.
- Use trusted websites.
- Avoid suspicious downloads.

- Do not install unknown software.
- Avoid entering passwords on suspicious websites.
- Use HTTPS websites when appropriate.
- Log out from shared computers.

On Public Computers

Always:

1. Log out of accounts.
2. Do not save passwords.
3. Clear downloaded sensitive files.
4. Avoid accessing highly sensitive accounts where possible.
5. Do not leave files behind.

Experiment 15: Safe Downloading

Objective: To learn how to download files safely.

Procedure

1. Search for a required document.
2. Prefer the official source.
3. Check the website/domain.
4. Download the file.
5. Check the file type.
6. Do not open unexpected executable files.
7. Scan suspicious files using appropriate security software.

Practice Example

Find a government notification PDF.

Students must identify:

- Website domain
- File name
- File type
- Publication date
- Whether the source appears official

Complete Internet and Cyber Hygiene Exercise

Objective: To combine all skills learned in previous tasks.

Scenario

You are a college student preparing a practical assignment on **Cyber Safety**.

Perform the following activities:

Task 1 — Search

Search the internet for reliable information about cyber hygiene.

Use:

"cyber hygiene" site:gov.in

Record two reliable sources.

Task 2 — Create Document

Create a document containing:

- Definition of cyber hygiene
- Five cyber threats
- Five safety practices

Task 3 — Cloud Storage

Create a Google Drive folder:

Cyber Hygiene Practical

Upload the document.

Task 4 — Sharing

Share the document with your instructor using appropriate permission.

Task 5 — Email

Compose an email to the instructor.

Include:

- Appropriate subject
- Short message
- Link/attachment to the assignment

Task 6 — Cyber Security

Before sending the email, verify:

- Correct recipient
- Correct attachment/link
- No confidential information
- Correct sharing permission

Task 7 — Phishing Identification

Analyze a sample suspicious email and identify at least five warning signs.

Task 8 — Password Security

Explain why password reuse is dangerous and describe how MFA improves security.

LAB ASSIGNMENT

Assignment 1

Create a Gmail account and demonstrate:

- Compose
- Send
- Reply
- Forward
- Cc
- Bcc
- Attachment
- Search
- Label
- Archive

Assignment 2

Create a Google Drive folder with:

Internet Tools
|

—	Notes
—	Assignments
—	Practical
—	Documents

Upload appropriate files.

Assignment 3

Prepare a one-page document titled:

"10 Rules of Cyber Hygiene"

Assignment 4

Find five phishing indicators from sample emails provided by the instructor.

Assignment 5

Demonstrate secure sharing of a Google Drive document.

VIVA-VOCE QUESTIONS

Basic Questions

1. What is the internet?
2. What is a web browser?
3. Give three examples of web browsers.
4. What is a URL?
5. What is a search engine?
6. What is Google Chrome?
7. What is Mozilla Firefox?
8. What is cloud storage?
9. What is Google Drive?
10. What is email?

Email Questions

11. What is Cc?
12. What is Bcc?
13. What is an email attachment?
14. What is phishing?
15. What is spam?
16. What is the difference between Reply and Reply All?

17. What is an email filter?
18. What is an email label?

Cyber Security Questions

19. What is cyber hygiene?
 20. What is malware?
 21. What is phishing?
 22. What is ransomware?
 23. What is social engineering?
 24. What is a strong password?
 25. Why should passwords not be reused?
 26. What is MFA?
 27. Why should OTPs not be shared?
 28. How can you identify a suspicious website?
 29. What should you do after receiving a suspicious email?
 30. Why should software and browsers be updated regularly?
-

LEARNING OUTCOMES

After completing this laboratory, students should be able to:

CO1 Skill	Expected Competency
Internet Browsing	Browse and navigate websites effectively
Search Techniques	Find relevant and reliable information
Email	Create and manage email communication
Cloud Storage	Upload, organize and share files
Online Communication	Participate safely in online meetings
Cyber Hygiene	Identify and avoid common cyber threats
Password Management	Create and manage secure passwords
Safe Internet Usage	Practice responsible and secure internet behavior